

Linux Workshop

ACM SIG Cyber

**Does anybody recognize this
logo ?**



Agenda

- Learn about what is linux
- Figure out which linux distribution to install
- Learn more about the linux distribution
 - Installing phase
 - Kali Linux Demo

What's Linux ?

- An open source OS (operating system)
- Linux has many distributions
- Every Linux distribution has the linux kernel
 - The Linux kernel manages all the hardware which makes up the whole OS
- Linux is very easy to manipulate which leads to the many different types of distributions



Beginner-friendly



Ubuntu
Based on Debian



Pop!_OS
Based on Ubuntu



elementary OS
Based on Ubuntu (LTS)



Mint
Based on Ubuntu



Zorin
Based on Ubuntu



Solus
[Independent]

Intermediate



Garuda Linux
Based on Arch



EndeavourOS
Based on Arch



Manjaro
Based on Arch



MX Linux
Based on Debian



Fedora
Based on Red Hat



OpenSUSE
[Independent]

Hard mode



Arch
[Independent] – DIY



Gentoo
[Independent] – DIY



Slackware
[Independent]



Linux From Scratch
[Independent] – DIY



Qubes OS
Based on Fedora – Security

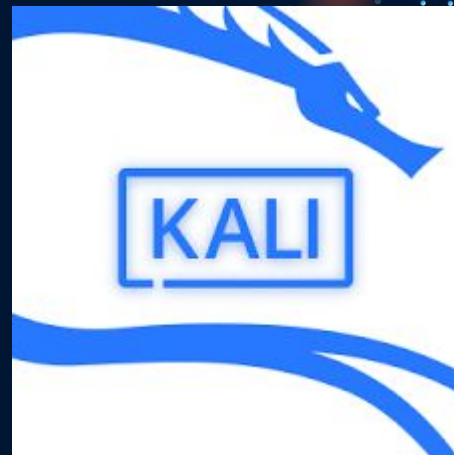
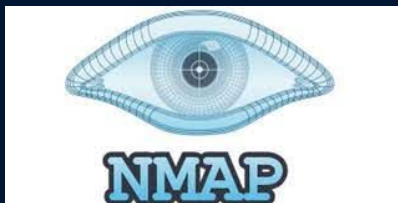


NixOS
[Independent] – DIY

Which Linux Distro are we going to use ?

We are going to be using a Linux Distro known as Kali Linux.

- Kali Linux is known for practicing ETHICAL hacking or pen testing.
- Kali has a diverse set of hacking tools
 - Nmap - network scanner
 - Aircrack-ng - packet sniffer



Kali Linux System Requirements

- Memory/RAM - 2 GB
- Disk space - 20 GB
- Processor - 64-bit Architecture



General Steps for Installing Kali Linux

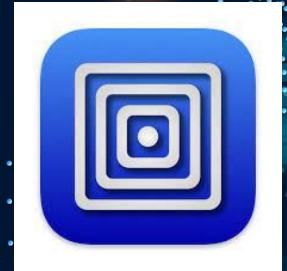
1. Install VMWare Player hypervisor
2. Download the Kali Linux pre-built Virtual Machine (VM)
3. Import the Kali Linux VM into VMWare Player
4. Start the Virtual Machine
5. Login with kali username and kali password
6. Begin using Kali linux tools



Install VM player

1. Visit the VMWare website to download the latest VMWare Workstation Player -
 - a. VMware is recommended for Windows systems
 - i. <https://www.vmware.com/go/getplayer-win>
 - b. UTM (QEMU) is recommended for Mac

UTM desktop icon



VMWare Workstation
Player desktop icon



Download the Kali Linux VM

1. Visit the Kali Linux website to download the software
 - a. <https://www.kali.org/get-kali/>
 - b. Direct download for VMWare VM - <https://cdimage.kali.org/kali-2023.3/kali-linux-2023.3-vmware-amd64.7z>
 - c. NOTE: Write down the default credentials: kali/kali
2. Click on the **Installer Images** graphic
3. Click on the **VMware 64-bit** graphic
4. The iso would be downloaded to the Downloads directory



Virtual Machines

- ✓ Snapshots functionality
- ✓ Isolated environment
- ✓ Customized Kali kernel
- ✗ Limited direct access to hardware
- ✗ Higher system requirements

VMware & VirtualBox pre-built images. Allowing for a Kali install without altering the host OS with additional features such as snapshots. Vagrant images for quick spin-up also available.

 Recommended

 Recommended

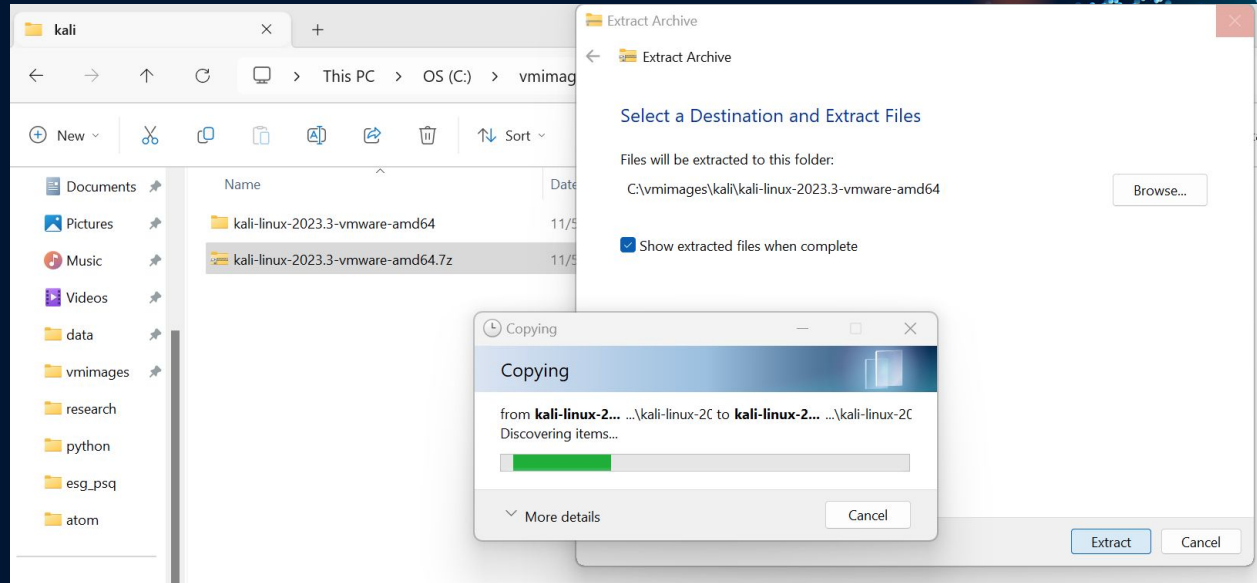


VMware

 2.9G	torrent	docs	sum
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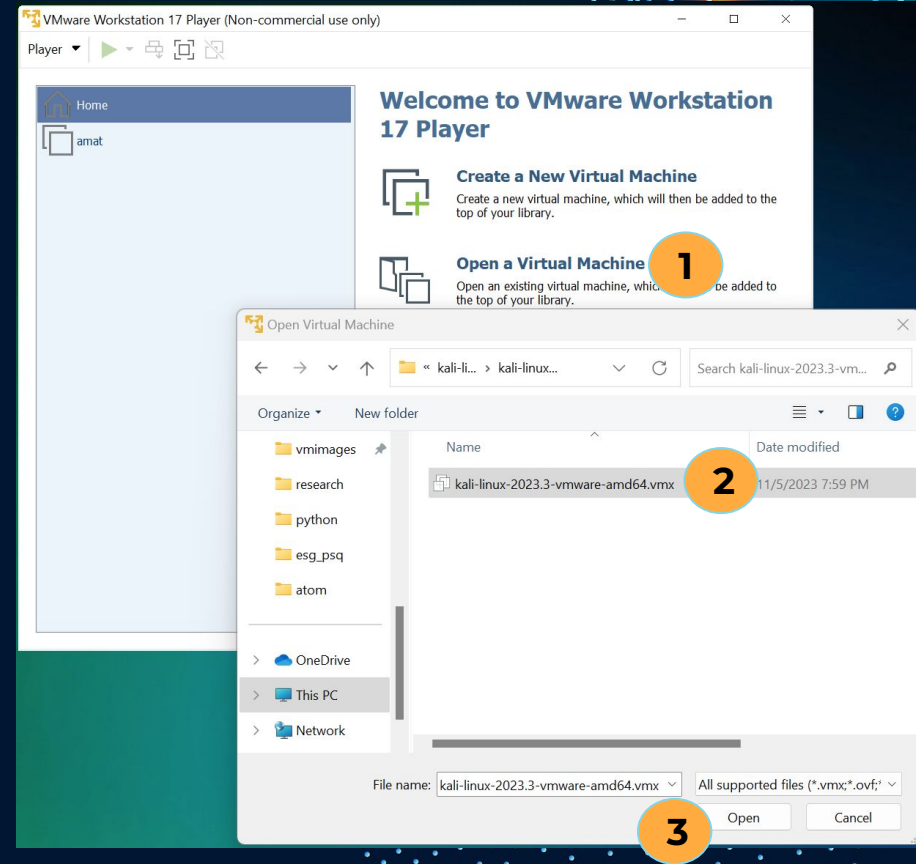
Extract the .7z VM image

1. Locate and select the downloaded VM image file **kali-linux-2023.3-vmware-amd64.7z**
2. Right-click and extract the file



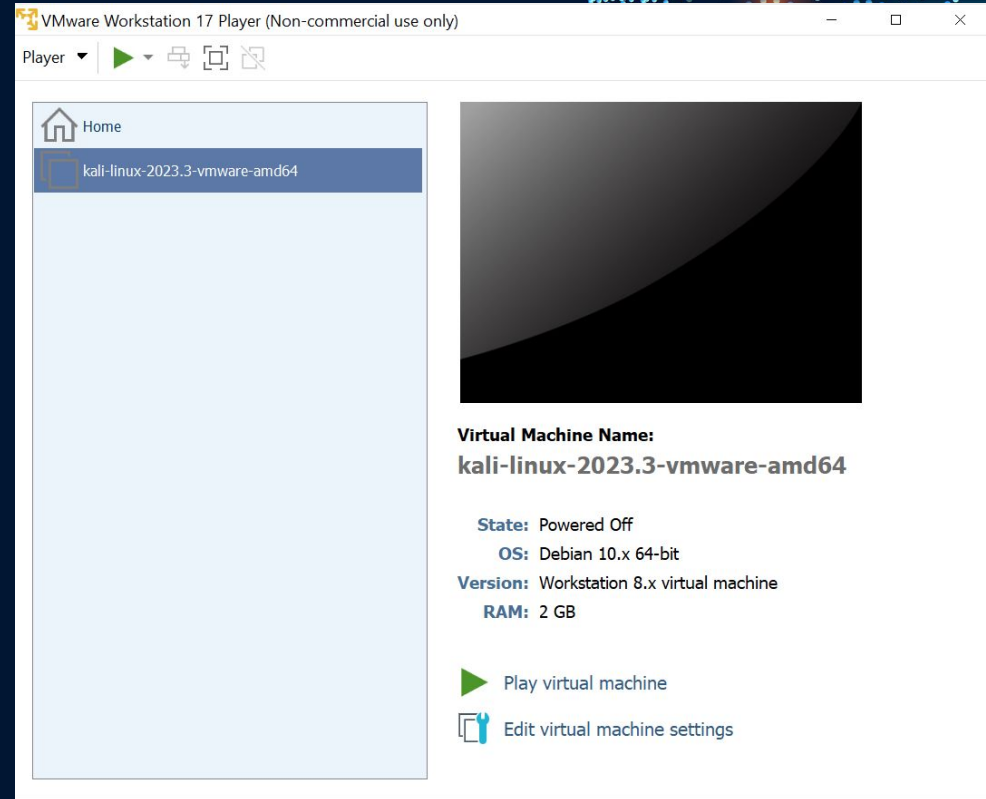
Importing VM into VMWare Player

1. Start VMware Player
2. Click on the **Open a Virtual Machine** link
3. Locate and select the extracted folder
 - a. Select the .vmx file
4. Click **Open** to import the VM



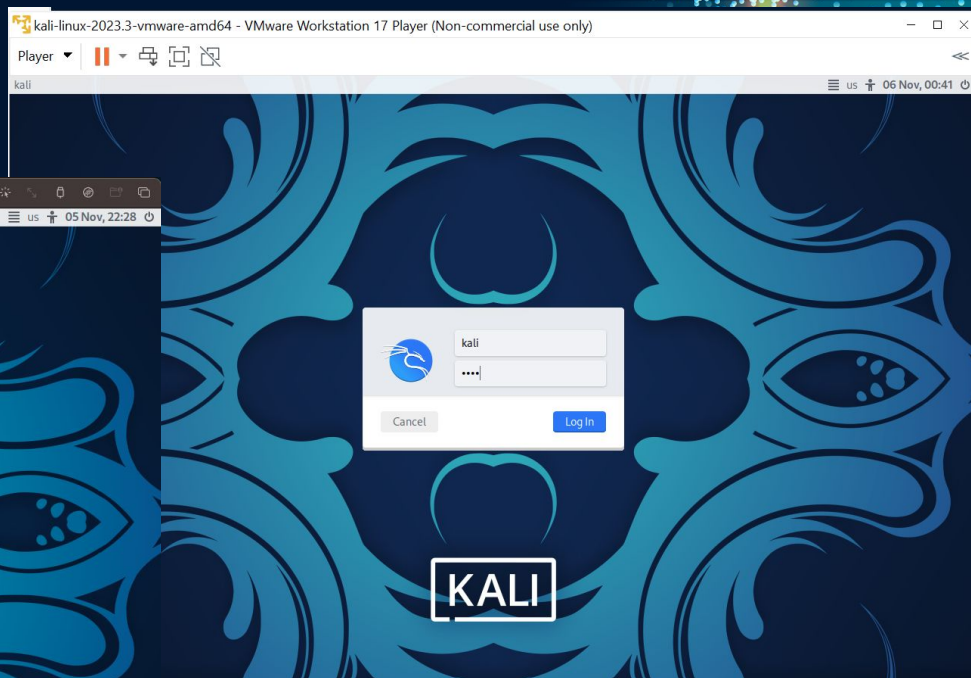
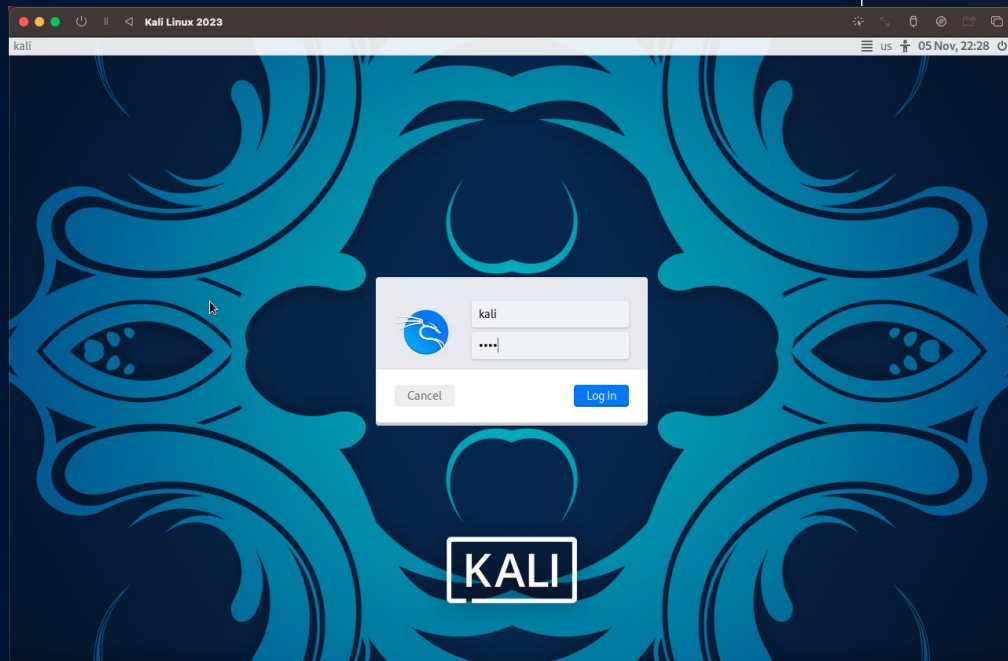
Starting the VM

1. Select the Kali Linux VM from the left side panel
2. Click on the **Play virtual machine** button to start the VM



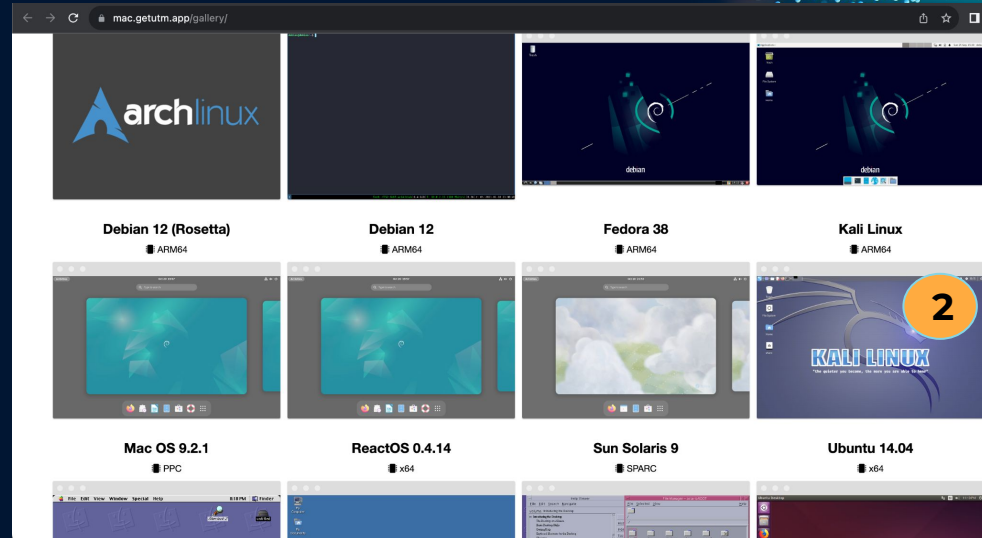
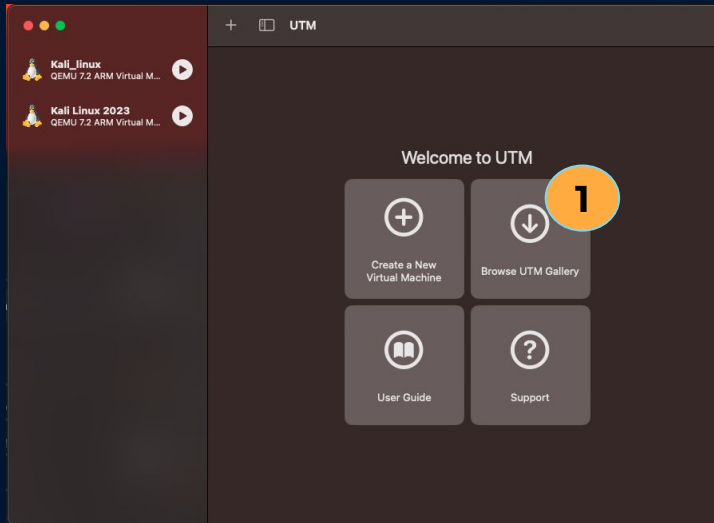
Login to Kali Linux VM

1. Enter the username: kali
2. Enter the password: kali

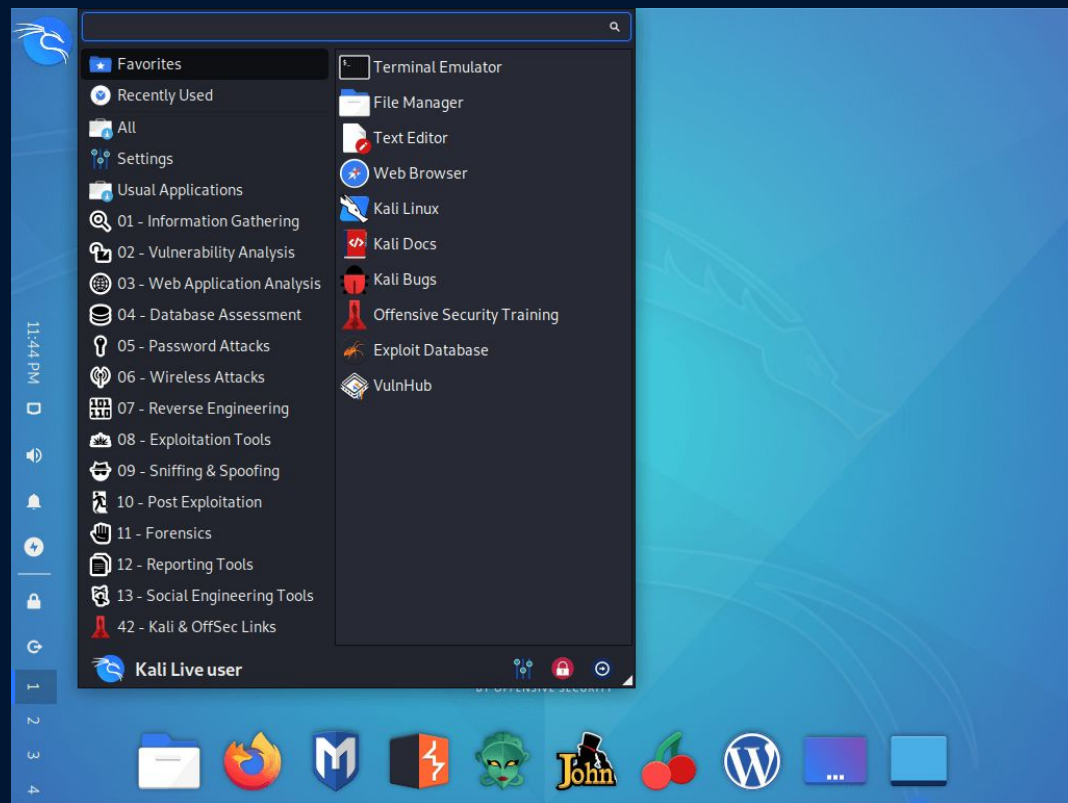


Downloading UTM prebuilt kali linux VM

- Upon opening the UTM app select Browse UTM Gallery
- It will send you to a webpage to choose from various pre built VMs
- Choose Kali linux



The Kali Linux Desktop



What's Hashcat

- Hashcat is a popular password cracking or “recovery” tool
- Passwords are usually stored as hashes on computer systems
- A hash is a way to encode a plain password into a short list of letters and/or numbers using an encryption algorithm

Hashcat can process a hashed password and try to crack it to get the original password



HASHCAT Demo

```
hashcat -m 0 -a 0 hash.txt passwordlist.txt
```

<https://infosecwriteups.com/cracking-hashes-with-hashcat-2b21c01c18ec>

```
-m 0
```

```
# Copy hash list file to the current directory
```

```
$ sudo cp /usr/share/wordlist/rockyou.txt.gz .
```

```
# Change ownership of the file to the akli user
```

```
$ sudo chown kali:kali rockyou.txt.gz
```

```
# Decompress rockyou.txt.gz
```

```
$ gzip -d rockyou.txt.gz
```

```
# Run hashcat against the rockyou.txt hash list
```

```
$ hashcat -m 0 -a 0 demo_hash.txt rockyou.txt
```



Thank You!!!!

my folder : `Downloads`

me : `cd downloads`

Linux :

